

C/C++ Implementation of Stack Operations

Stack Implementation Using Array

```
#include <stdio.h>
#include <stdlib.h>
#define MAX 10

void push();
int pop();
void display();
int stack[MAX];
int top = -1;

void main()
{
    int ch;
    char ch1;
    do
    {
        clrscr();
        printf("\n 1 -> Push");
        printf("\n 2 -> pop");
        printf("\n 3 -> Display");
        printf("\nEnter your choice");
        scanf("%d", &ch);
        switch(ch)
        {
            case 1: push();
                    break;
            case 2: printf("The deleted item is %d", pop);
                    break;
        }
    }
}
```


case 3: **display()**;

break;

default: printf("you have entered wrong choice\n");

printf("do you want to continue (Y/N)");

scanf(" %c", &ch1);

}

while (ch1 == 'y' || ch1 == 'Y');

}

void push()

{
int item;

if (top == MAX - 1)

{
printf("overflow\n");

exit(0);

}

else

{

printf("Enter the element which you want to insert");

scanf(" %d", &item);

top = top + 1;

stack[top] = item;

}

}


```
int pop()
```

```
{
```

```
    int i;
```

```
    if (top == -1)
```

```
    {
```

```
        printf("\n Underflow condition");
```

```
        exit(0);
```

```
    }
```

```
    else
```

```
    { i = stack[top];
```

```
      top = top - 1;
```

```
    }
```

```
    return(i);
```

```
}
```

```
void display()
```

```
{
```

```
    int i;
```

```
    if (top == -1)
```

```
    { printf("\n stack is empty");
```

```
      exit(0);
```

```
    }
```

```
    else
```

```
    {
```

```
        for (i = top; i >= 0; i--)
```

```
        { printf("\n the item is %d", stack[i]);
```

```
        }
```

```
    }
```

```
}
```